

Vishal Kumar

+1-213-421-6583 — vishalku@usc.edu

GitHub — LinkedIn

EDUCATION

University of Southern California	2025 – 2027 (<i>Expected</i>)
M.Tech. in Computer Science	
Guru Gobind Singh Indraprastha University	2020 – 2024
B.Tech. in Computer Science, CGPA: 8.7	
CBSE Board	2020
Senior Secondary, CGPA: 9.3	

EXPERIENCE

Teaching Assistant, USC Viterbi K-12 STEM Center	Jun 2025 – Aug 2025
<i>University of Southern California</i>	
Taught Cybersecurity , Python , and Game Development to high school students as part of USC's outreach program. Designed and delivered interactive lessons that simplified complex CS concepts and encouraged hands-on learning. Fostered an inclusive environment that promoted curiosity, creativity, and foundational technical skills.	
Software Developer Intern	Jun 2024 – Sep 2024
<i>Punjab National Bank</i>	
Collaborated with senior developers to design and implement large-scale digital banking solutions for a government-owned financial institution. Contributed to development, testing, and enhancement of customer-facing applications; gained practical experience in agile development , secure software design , and enterprise-grade systems.	

RESEARCH PUBLICATIONS

Generative A.I. Integration using APIs
Authored under Dr. Sangeeta and Dr. Sitender at Maharaja Surajmal Institute of Technology.
Submitted to <i>Satyam</i> (Nov 2023). Explores the synergy between Generative A.I. and APIs , providing insights, integration frameworks, and practical implementation examples.

PROJECTS

Taxi Demand Prediction	<i>Python, Dask, ARIMA, LSTM, Data Analysis, Time Series Forecasting</i>
Developed a robust machine learning model to predict NYC taxi demand using 2017–2018 datasets. Performed comprehensive EDA and feature engineering ; implemented ARIMA and LSTM models for real-time forecasting; optimized driver positioning to reduce idle time and improve operational efficiency.	
Malaria Classification System	<i>Python, CNN, Deep Learning, Data Augmentation</i>
Built a CNN-based malaria detection system using medical imaging data. Preprocessed and augmented datasets to improve generalization, achieving 90% classification accuracy . Automated deep learning pipelines to accelerate diagnostic workflows and improve healthcare accessibility.	
Apple iPhone Website Clone	<i>React.js, Three.js, GSAP, JavaScript, Vite</i>
Created a responsive Apple website clone showcasing interactive 3D modeling and animations. Integrated Three.js for graphics and GSAP for motion design; leveraged React hooks, Context API, and modular styling. Deployed with Vite for a high-performance, production-ready experience.	

CERTIFICATIONS

iOS Mobile Application Development (Meta)
Docker Essentials (IBM)
Cloud Computing (AcmeGrade)
MERN Stack (Brain Mentors)
Python for Data Science (IBM)

SKILLS

Programming Languages: Python, Swift, JavaScript, MySQL, C#
Frameworks: React.js, Node.js, SwiftUI, Next.js
Developer Tools: Git, PostgreSQL, Microsoft Azure, Unity, GitHub, Docker, Xcode
Core Skills: Web Development, Data Science, Machine Learning, AI, Deep Learning, Game Development